Introduction to Finance for Data Scientists

Session 5: Bond Markets

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Introduction

- Fixed income securities give the owner a predetermined cash flow, typically a bond.
- Average daily U.S. trading volume 2024, according to SIFMA:
  - Equities: 607 bn USD/day.
  - Fixed income: 1290 bn USD/day.
- Key event on this market is discrete and rare: default of the borrower.
- Data and methodologies quite different from equity markets. Good example of the use of structural models.

Road map

Overview of fixed income markets
- Vocabulary
- Market functioning

Forecasting credit losses
- Credit Ratings
- Merton's Model - Derivation
- Merton's Model - Implications

Discussion on Structural Models

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Example - Bonds - Boeing, 1 May 2020



The Boeing Company

$3,000,000,000 4.508% Senior Notes due 2023

$3,500,000,000 4.875% Senior Notes due 2025

$2,000,000,000 5.040% Senior Notes due 2027

$4,500,000,000 5.150% Senior Notes due 2030

$3,000,000,000 5.705% Senior Notes due 2040

$5,500,000,000 5.805% Senior Notes due 2050

$3,500,000,000 5.930% Senior Notes due 2060

We are offering $3,000,000,000 aggregate principal amount of our 4.508% senior notes due 2023 (the "2023 notes"), $3,500,000,000 aggregate principal amount of our 4.875% senior notes due 2025 (the "2025 notes"), $2,000,000,000 aggregate principal amount of our 5.040% senior notes due 2027 (the "2027 notes"), $4,500,000,000 aggregate principal amount of our 5.150% senior notes due 2030 (the "2030 notes"), $3,000,000,000 aggregate principal amount of our 5.705% senior notes due 2040 (the "2040 notes"), $5,500,000,000 aggregate principal amount of our 5.805% senior notes due 2050 (the "2050 notes") and $3,500,000,000 aggregate principal amount of our 5.930% senior notes due 2060 (the "2060 notes" and, together with the 2023 notes, the 2025 notes, the 2027 notes, the 2030 notes, the 2040 notes and the 2050 notes, the "notes").

The 2023 notes will mature on May 1, 2023. The 2025 notes will mature on May 1, 2025. The 2027 notes will mature on May 1, 2027. The 2030 notes will mature on May 1, 2030. The 2040 notes will mature on May 1, 2040. The 2050 notes will mature on May 1 , 2050. The 2060 notes will mature on May 1, 2060.

We will pay interest on the notes on each May 1 and November 1, commencing on November 1, 2020.

We may redeem the notes prior to maturity, in whole or in part, at the respective redemption prices set forth herein. See "Description of Notes—Optional Redemption." The notes will not be listed on any securities exchange. Currently, there are no public markets for the notes.

The notes will be our unsecured senior obligations. The notes will rank equally in right of payment with all of our existing and future unsecured and unsubordinated indebtedness and will rank senior in right of payment to any existing and future indebtedness that is subordinated to the notes.

Basic definitions

- The borrower gets an amount M today.
- It promises to repay $D = (1 + r)M$ at a future date T.
- Some vocabulary:
  - M is the principal.
  - r is the interest rate, rM are interests.
  - D is the nominal value (also face value, par value) of the bond.
  - T is the maturity of the bond.
- Notice the difference with a stock: the contract specifies D ("fixed income").
- There are many different products with variations on these features.

Contractual variations

- r can:
  - Have a fixed value: fixed rate.
  - Vary with inflation/benchmark rates/exchange rates according to a deterministic formula: variable rate.
- Timing of repayments:
  - Principal and interests repaid at maturity only.
  - Interests paid periodically via coupons, principal repaid at maturity.
  - Principal and interests both repaid gradually.

Different counterparties

- Different borrowers:
  - Corporate debt.
  - Government debt.
  - Household debt: Mortgages, credit card debt, student loans...
- Different negotiability:
  - Negotiable: bonds.
  - Non-negotiable: (bank) loans.

Different maturities

- Short maturity: the money market, debt instruments with maturity less than 1 year.
- For U.S. Treasury securities:
  - T-Bills: Less than 1 year.
  - T-Notes: 2-10 years.
  - T-Bonds: 30 years.
- Infinite maturity: perpetual bonds ("perp").

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Participants

- Fixed income securities diverse and highly customized.
- Imagine you want to buy "a" Deutsche Bank bond.
  Well, 168 different ones are traded on Boerse Frankfurt only.
- Highly customized product makes it difficult to trade on an open exchange.
- Most fixed income markets are dealer markets and over the counter.
- This means that each transaction has a different price, and many prices are not made public.

Name ▲▼	WKN	Last price ▲▼	Date/Timestamp last price▲▼	Turnover in Euro▼	+/- % ▲▼
Deutsche Bank AG 0,000% 13/43	A1TNFQ	18.00	01/10/25 11:13	5,016	2.86
Deutsche Bank AG 0,000% 96/26	134224	96.75	01/10/25 12:58	0	0.00
Deutsche Bank AG 2,287% 11/26	A1C956	99.95	01/10/25 12:09	0	-0.05
Deutsche Bank AG 1,05% 21/31	A3MP40	88.93	01/10/25 11:26	0	0.10
Deutsche Bank AG 2,475% 25/32	A4DFJL	100.06	01/10/25 11:23	0	0.00
Deutsche Bank AG 3,52% 25/45	A4DFLL	99.00	01/10/25 12:09	0	0.00
Deutsche Bank AG 2,625% 25/28	A4DFSK	99.70	01/10/25 09:22	0	-0.03
Deutsche Bank AG 3,84% 25/55	A4DFSU	98.99	01/10/25 11:23	0	-0.37
Deutsche Bank AG 3,51% 25/33	A4DFUD	98.48	01/10/25 12:36	0	0.00
Deutsche Bank AG 3,78% 25/55	A4DFWB	99.00	01/10/25 12:09	0	0.00

Issuance and prices

- Consider a new bond with a par value 100 USD.
- At issuance the bond is sold by dealers to their customers at $P_0 < 100$.
- The return on this bond at issuance is $y_0 = \frac{100}{P_0} - 1$: the bond's yield.
- The price of the bond will move over time, so does the yield: $y_t = \frac{100}{P_t} - 1$.
- Simple but important: the yield goes down means that the price goes up.

Determinants of prices

- r_t the interest rate on 3-months T-bills is considered the risk-free rate (in USD).
- The difference $y_t - r_t$ is called the yield spread: additional compensation required by an investor to hold a bond rather than risk-free debt.
- The yield spread primarily reflects credit risk.
- For a given borrower, if $y_{t,T}$ is yield at time t for a bond at maturity T, one can look at $\{y_{t,T}\}$ for all maturities T:
  the term structure of interest rates, or yield curve.
- The yield curve primarily reflects inflation expectations and interest rate risk.

How do we price credit risk?

- Main question to ask: what if the borrower doesn't repay?
- When a borrower misses a payment, it defaults on a debt contract.
- Ex. 2020 Sovereigns: Argentina, Lebanon.
- Ex. 2020 Corporates: 14.6 bn USD defaults in the U.S. in May alone.

Source: Giesecke, Longstaff, Schaefer, and Strebulaev (2011).

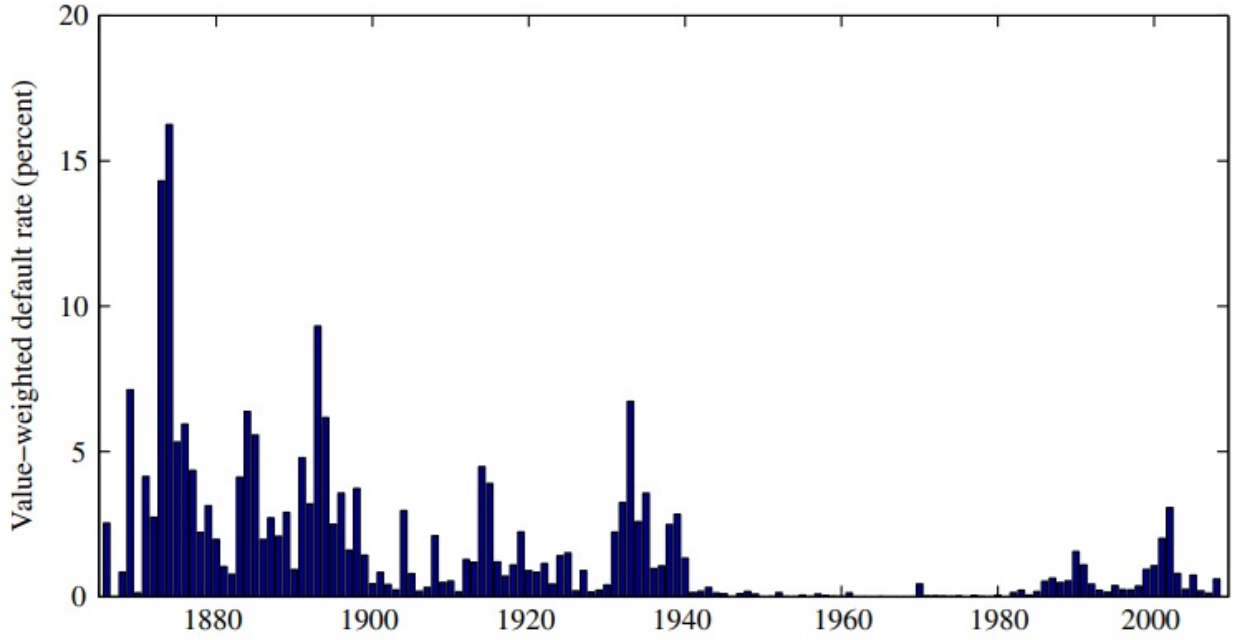


Fig. 1. Historical default rates. This graph plots the annual value-weighted percentage default rates for bonds issued by U.S. domestic nonfinancial firms for the 1866–2008 period.

Vocabulary and sophisticated maths

- Lenders care first about the probability of default (PD).
- The amount on which the borrower defaults is called exposure at default (EAD).
- If after liquidation/restructuring the lender recovers R, R/EAD is called the recovery rate.
- Conversely, 1 minus the expected recovery rate is the losses given default (LGD).
- Finally, the expected losses (EL) are defined as:

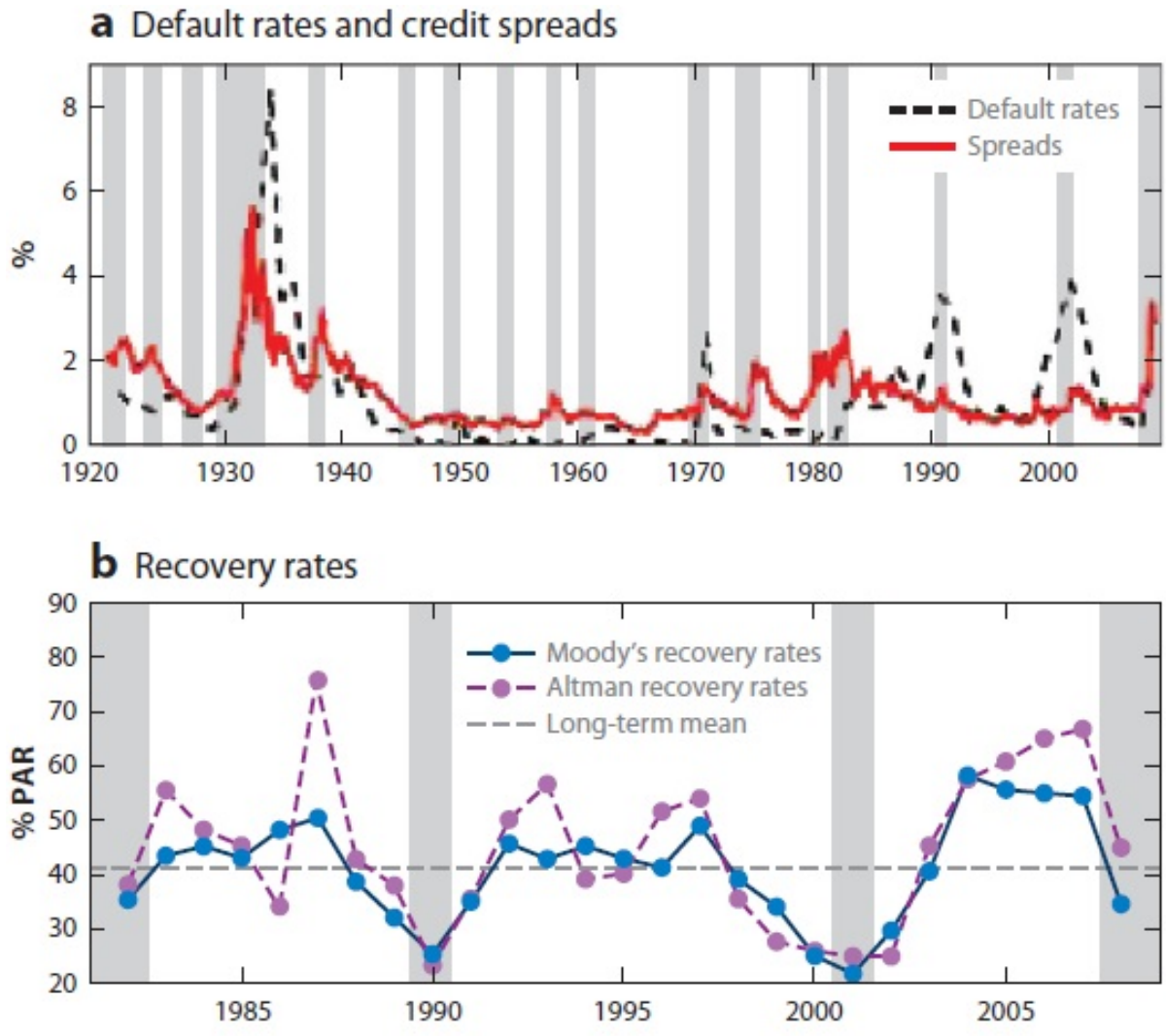
$$EL = PD \times EAD \times LGD$$

How are LGD determined?

What happens after a default?

- Option 1: Liquidation.
  - Assets sold at best possible price, as a whole or piecewise.
  - Proceeds used to reimburse creditors in a predetermined order. The rank of each creditor is called seniority.
- Option 2: Restructuring.
  - Shareholders and creditors agree on a solution that makes everyone better off.
  - Typically: rescheduling/write-down of debt, debt for equity swap...
  - Legal institutions are very important to determine the outcome.

Data from Chen (2010)



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Credit Rating Agencies

- Bond issuers pay Credit Ratings Agencies to issue a rating.
- This rating is a (qualitative) view on the issuer's creditworthiness.
- "Big Three": S&P Global Ratings, Moody's, Fitch Group.

Transition Matrix

Source: FitchRatings.

Fitch Global Corporate Finance Transition Matrices: 1990–2024

Average Annual

No.	(%)	AAA	AA	A	BBB	BB	B	CCC to C	D	WD
942	AAA	88,75	5,20	0,21	-	-	-	-	0,11	5,73
6 601	AA	0,12	86,70	8,29	0,30	0,02	0,02	-	0,05	4,51
22 175	A	0,00	1,43	89,48	4,65	0,31	0,04	0,03	0,05	4,01
26 836	BBB	0,00	0,09	2,65	88,32	3,00	0,29	0,08	0,12	5,45
11 733	BB	-	0,02	0,08	6,07	78,79	5,44	0,95	0,58	8,07
8 991	B	-	-	0,13	0,21	6,82	75,98	4,57	2,02	10,27
1 412	CCC to C	-	-	-	0,14	1,13	14,59	48,23	23,87	12,04

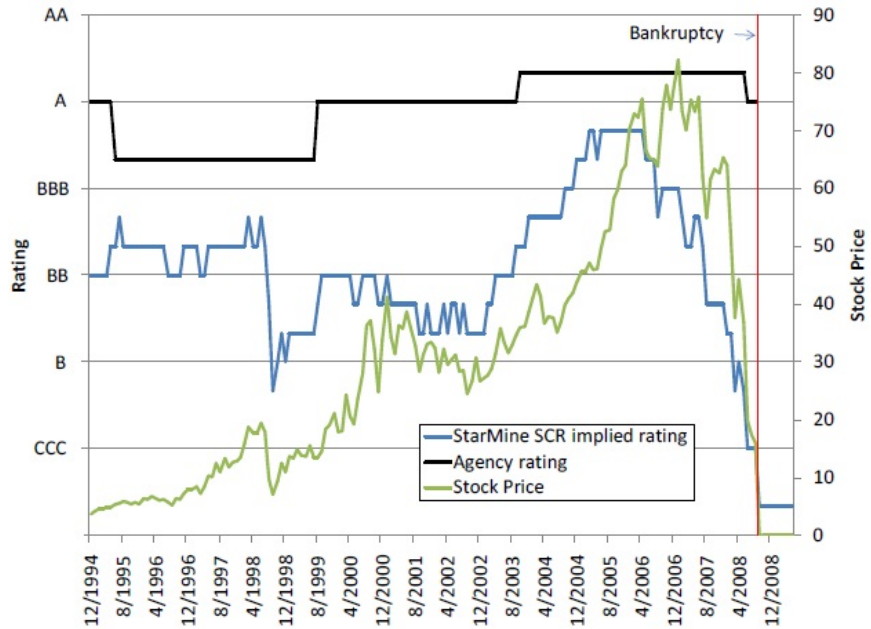
Impact of Ratings

- Bonds with grade BBB- (S&P, Fitch) or Baaa3 (Moody's) or below are called junk / speculative-grade / non-investment-grade bonds.
- Being riskier, they command a higher yield and are called high-yield bonds.
- Other bonds are called investment grade bonds.
- Many funds (in particular pension funds) can only invest in investment grade bonds.
- Being downgraded to junk can have a huge impact for a company.

Why are ratings not enough?

- CRAs have access to privileged corporate information.
- However, ratings have a very low frequency.
- Market prices incorporate more information, more quickly.

From Bonne and Malinak (2013) - Lehman Brothers.



Some thoughts

- Stock price seems to forecast default.
- Indeed, denote:
  - V_t the market value of a company's assets.
  - B_t the market value of its debt.
  - E_t the market value of its stock.
- We have:

$$V_t = B_t + E_t$$

- Stock price gives E_t, informative about V_t and hence B_t.

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Merton (1974)

- Game changer in the analysis of credit risk;
- Idea: stock price gives a lot of information on the risk of a company's assets.
- This information can be extracted and used to evaluate the credit risk of that company.
- This is achieved through a "structural model" giving a theoretical relationship between stock prices and debt repayment.

Setup

- A company has debt D to repay at time T. Only stocks and bonds.
- It operates assets with market value V_t at time t. V_t follows a geometric brownian motion:

$$\frac{dV_t}{dt} = \mu V_t + \sigma V_t \frac{dz}{dt}, \text{ with } \frac{dz}{dt} \hookrightarrow \mathcal{N}(0,1).$$

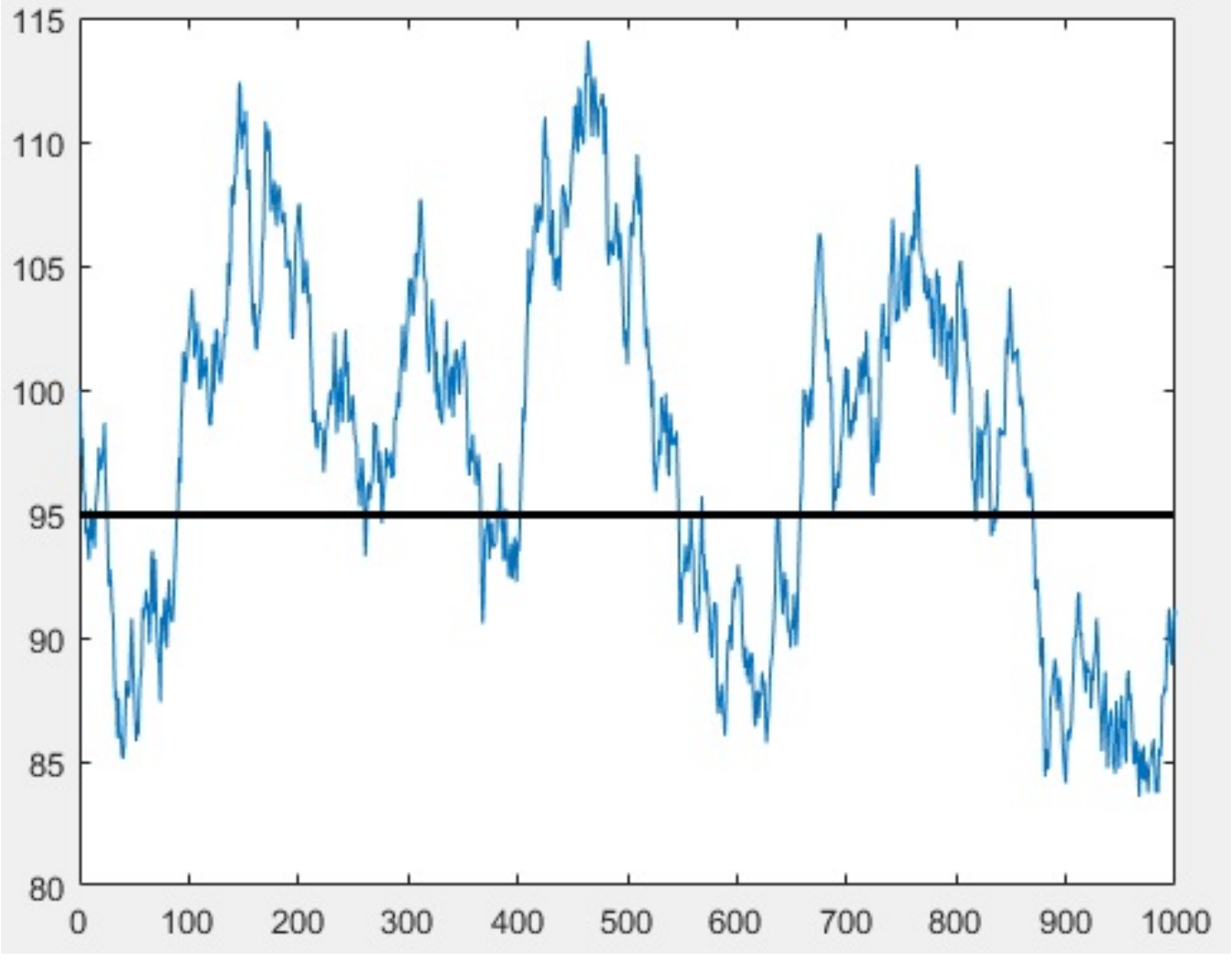
- The company will default if at time T we have $V_T < D$.
- Denote E_t the value of the company's equity, B_t the value of the company's bonds.

$$\begin{aligned} E_T &= \max(V_T - D, 0) \\ B_T &= \min(D, V_t) \\ \forall t \leq T,\ V_t &= E_t + B_t \end{aligned}$$

- Question: what is the value of B_t, i.e., the price of the bond?

Example

Evolution of V_t. $V_0 = 100$ and $D = 95$.



Tool: Ito's lemma

If $\frac{dz}{dt} \hookrightarrow \mathcal{N}(0,1)$ and x follows:

$$\frac{dx}{dt} = a(x,t) + b(x,t)\frac{dz}{dt}$$

Then for any function $f(x,t)$ twice differentiable we have:

$$\frac{df}{dt} = \left(\frac{\partial f}{\partial x}a + \frac{\partial f}{\partial t} + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}b^2\right) + \frac{\partial f}{\partial x}b\frac{dz}{dt}.$$

(intuitively, these are the first terms of the Taylor expansion of $f(x,t)$)

Applying Ito's lemma

- The bond price B_t is a function of V_t and t: $B_t = B(V_t, t)$.
- Assume B is twice-differentiable (to be checked).
- Then Ito's lemma tells us:

$$\frac{dB}{dt} = \left(\frac{\partial B}{\partial V} \mu V + \frac{\partial B}{\partial t} + \frac{1}{2} \frac{\partial^2 B}{\partial V^2} \sigma^2 V^2 \right) + \frac{\partial B}{\partial V} \sigma V \frac{dz}{dt}.$$

The replication portfolio

- Imagine on any interval $[t, t+dt]$ I'm long 1 bond, and short $\frac{\partial B}{\partial V} / \left(1 - \frac{\partial B}{\partial V}\right)$ the stock.
- Using $V_t = B_t + E_t$, the value of this portfolio at time t is:

$$\Pi_t = B_t - \frac{\frac{\partial B}{\partial V}}{1 - \frac{\partial B}{\partial V}} E_t = \frac{1}{1 - \frac{\partial B}{\partial V}} \left(B_t - \frac{\partial B}{\partial V} V_t \right).$$

- Π_t evolves as:

$$\frac{d\Pi_t}{dt} = \frac{1}{1 - \frac{\partial B}{\partial V}} \left(\frac{dB_t}{dt} - \frac{\partial B}{\partial V} \frac{dV_t}{dt} \right)$$

- Using the expression for $\frac{dB}{dt}$ above and replacing one gets:

$$\frac{d\Pi_t}{dt} = \frac{1}{1 - \frac{\partial B}{\partial V}} \left(\frac{\partial B}{\partial t} + \frac{1}{2} \frac{\partial^2 B}{\partial V^2} \sigma^2 V^2 \right)$$

No arbitrage condition

- The portfolio's evolution is independent of $\frac{dz}{dt}$: no risk!
- Such a portfolio is called "delta-hedged".
- If the portfolio is risk-free, it needs to have the same return as the risk-free asset (Session 2).
- Otherwise there would be an arbitrage. How?
- This gives the no-arbitrage condition:

$$\begin{aligned} r\Pi_t &= \frac{d\Pi}{dt} \\ \Leftrightarrow rB_t &= \frac{\partial B}{\partial t} + rV_t\frac{\partial B}{\partial V} + \frac{1}{2}\frac{\partial^2 B}{\partial V^2}\sigma^2 V_t^2 \end{aligned}$$

Solution

- We thus have the Black-Scholes-Merton partial differential equation:

$$rB_t = \frac{\partial B}{\partial t} + rV_t\frac{\partial B}{\partial V} + \frac{1}{2}\frac{\partial^2 B}{\partial V^2}\sigma^2 V_t^2$$

- In addition, we have the following terminal condition:

$$B_T = \min(V_T, D)$$

- A solution is given by:

$$\begin{aligned} B_t &= V_t - V_t N(d_1) + De^{-r(T-t)}N(d_2) \\ \text{with } d_1 &= \frac{\ln(V_t/D) + (r+\sigma^2/2)(T-t)}{\sigma\sqrt{T-t}} \\ d_2 &= d_1 - \sigma\sqrt{T-t} \\ N & \quad \text{cdf of } \mathcal{N}(0,1) \end{aligned}$$

- The probability of default is given by $PD = N(d_2)$.

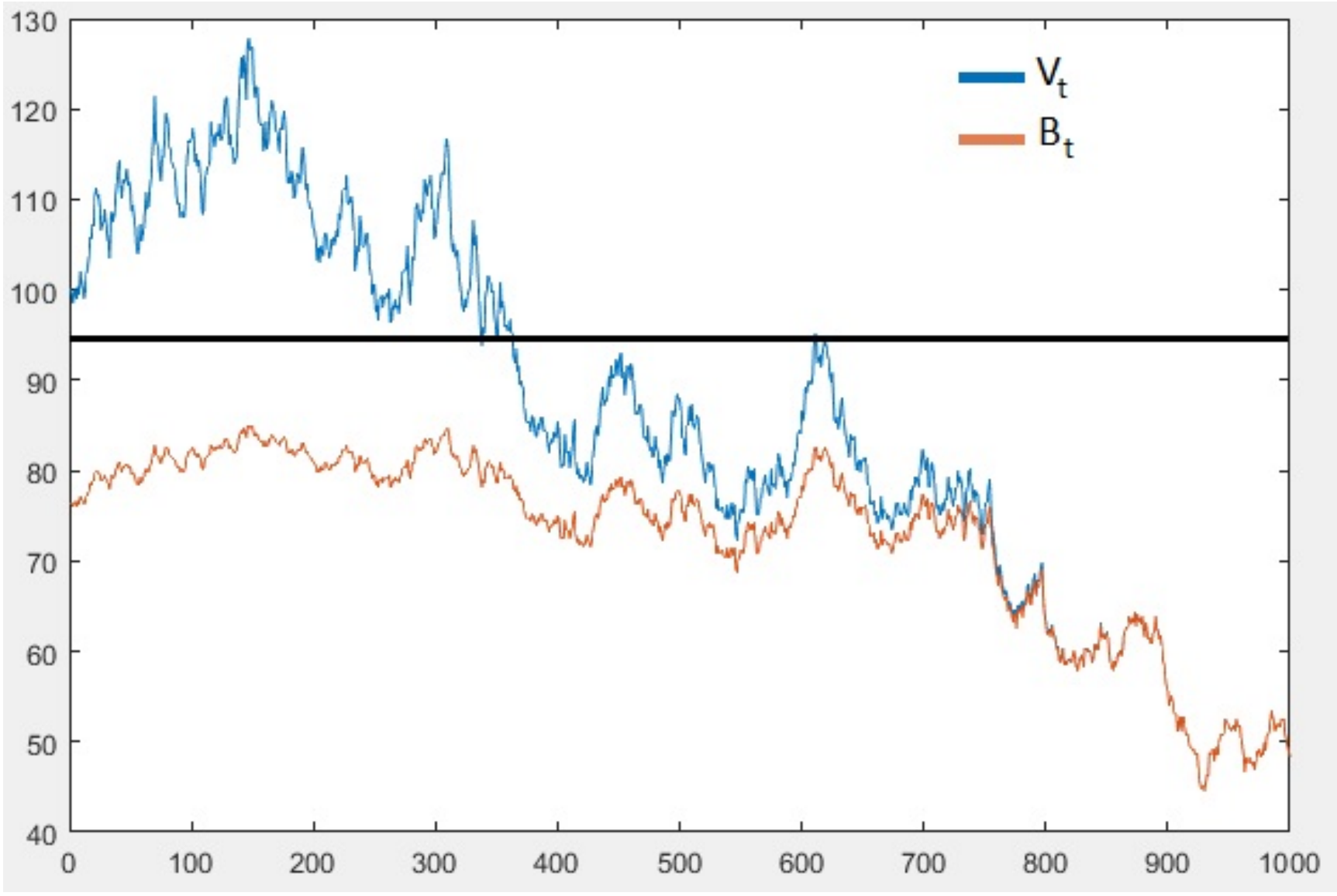
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Simulation


130
120
110
100
90
80
70
60
50
40
0
100
200
300
400
500
600
700
800
900
1000
V_t
B_t

Using the formula on data

- V_t and σ are not directly observable.
- Denote σ_E the standard deviation of E_t (stock prices, observable).
- The model yields the following two equations:

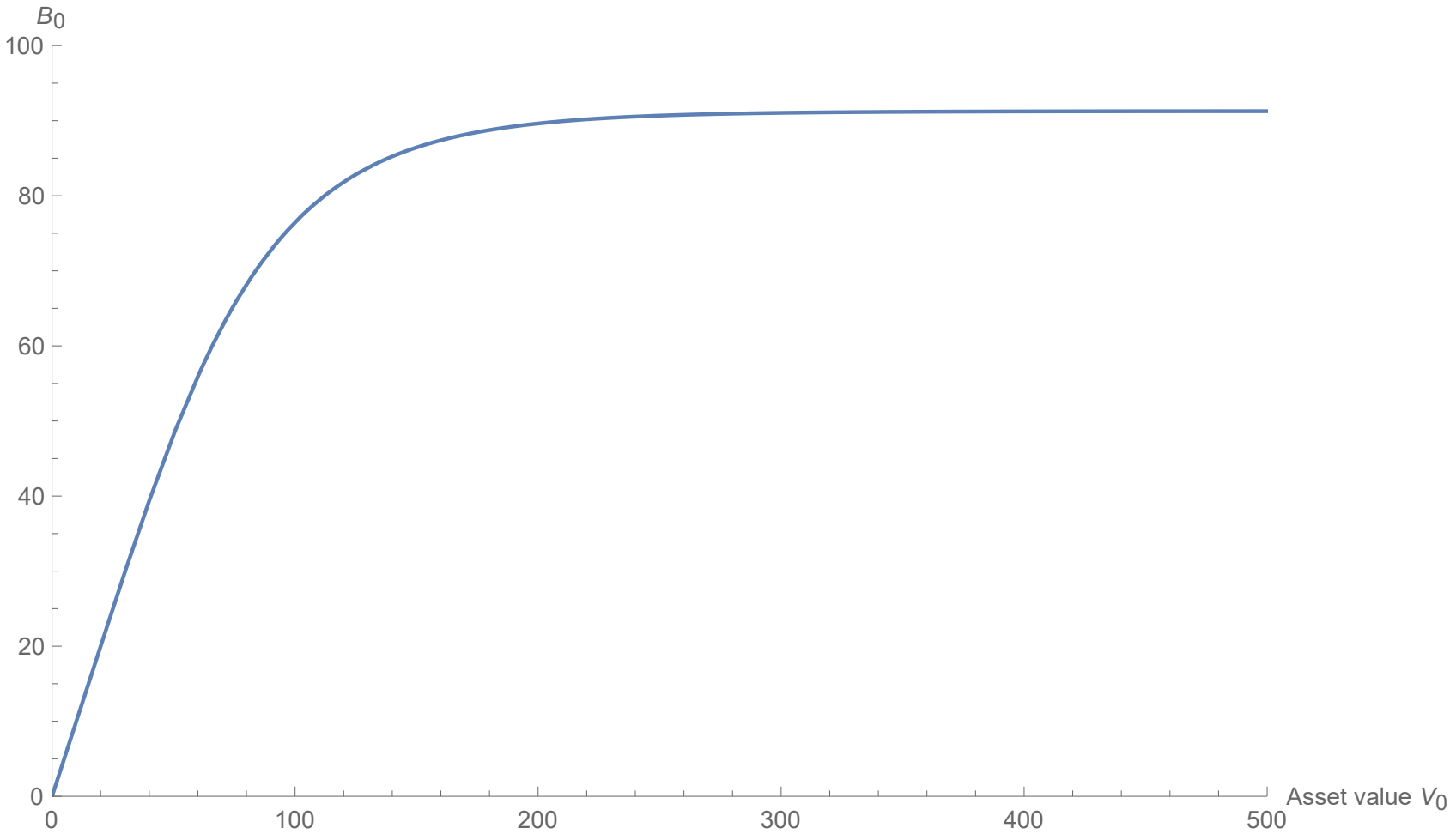
$$\begin{aligned} \sigma_E E_0 &= N(d_1)\sigma V_0 \\ E_0 &= V_0 N(d_1) - De^{-rT} N(d_2) \end{aligned}$$

- Observing E_0 and σ_E in the data, one can solve for V_0 and σ_V.
- We then have all the parameters of the model.
- More on this topic: John Hull, Options, Futures, and Other Derivatives, 2015.

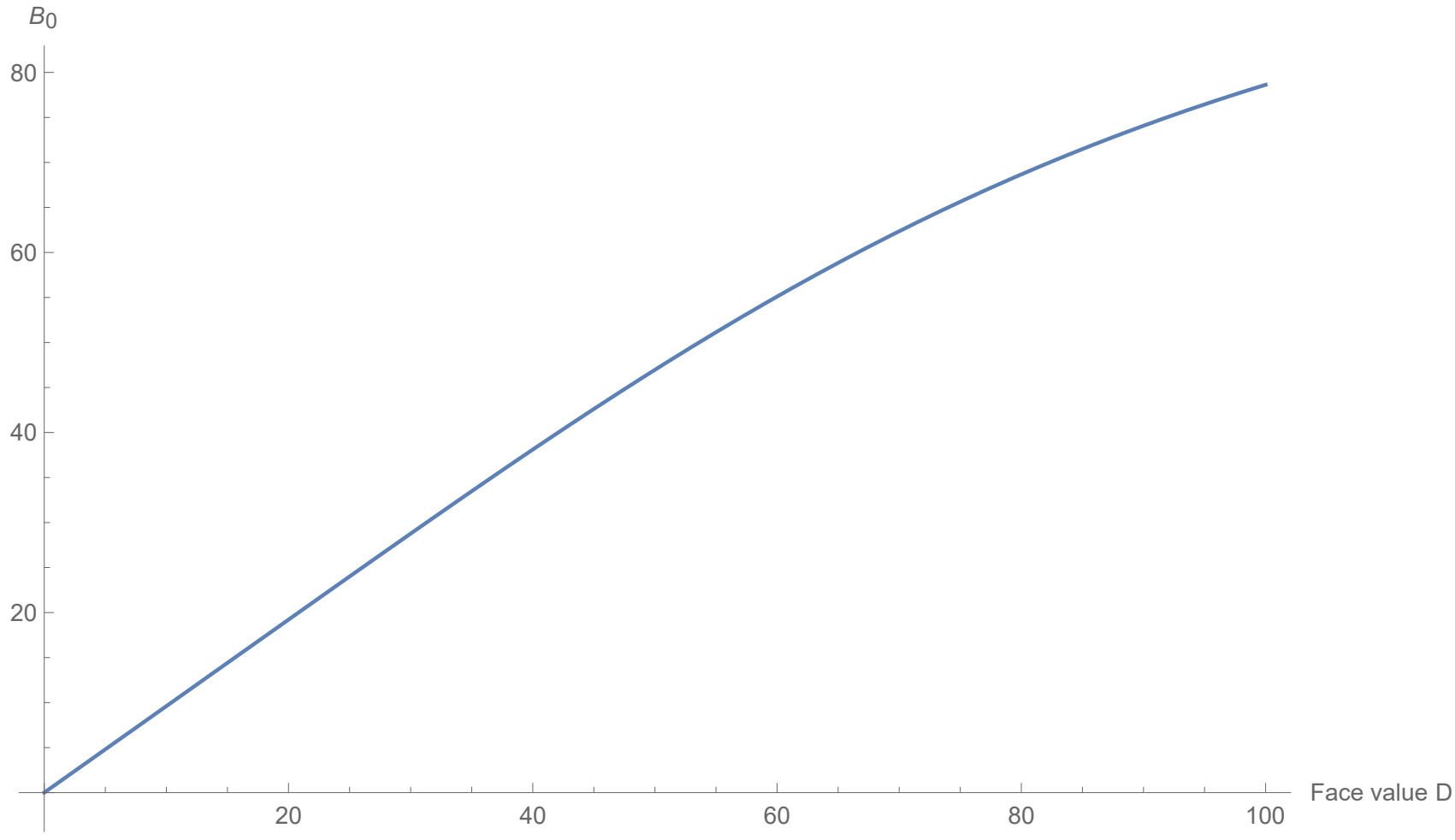
Comparative statics

Can you propose an intuition for each of the following graphs?

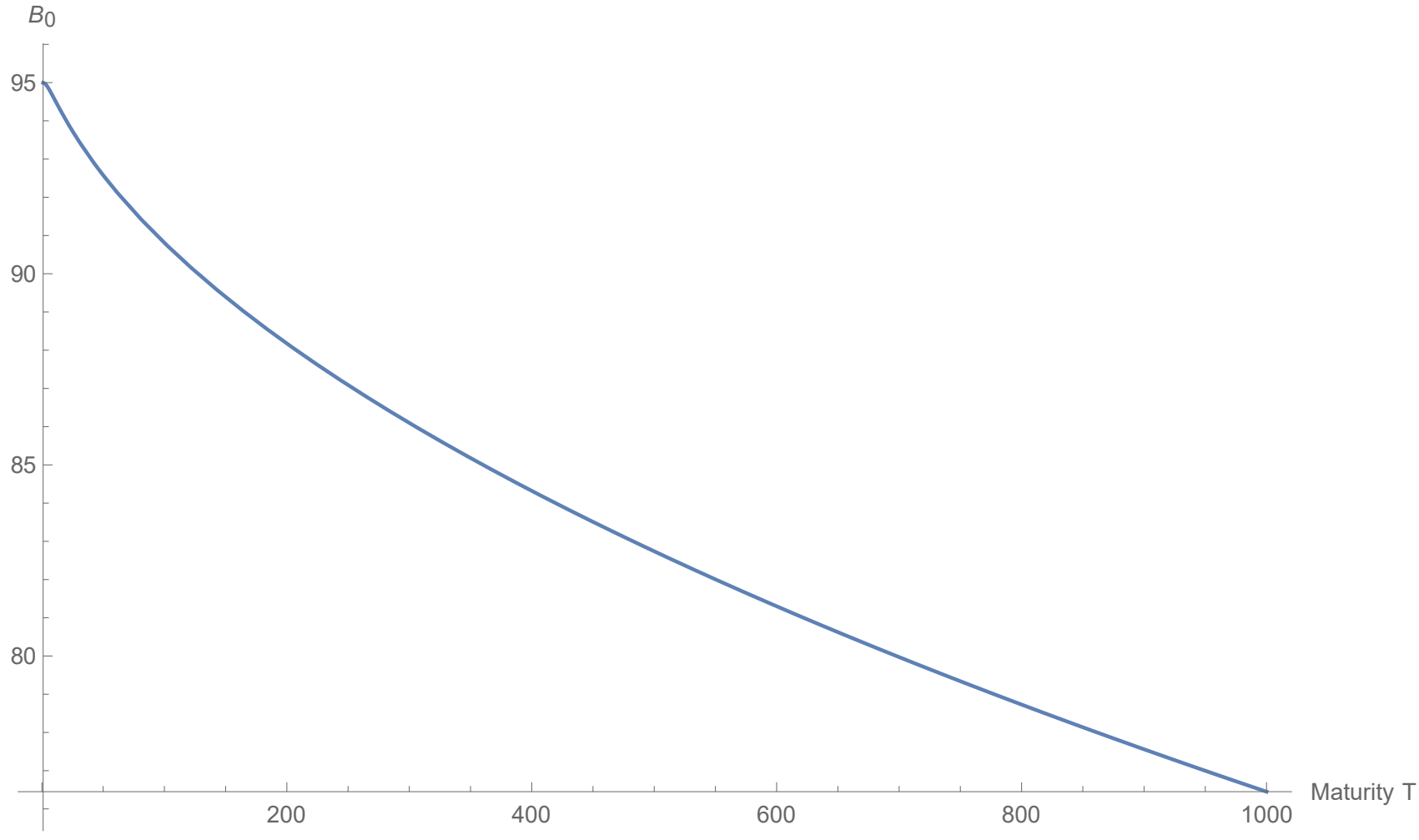
Asset value



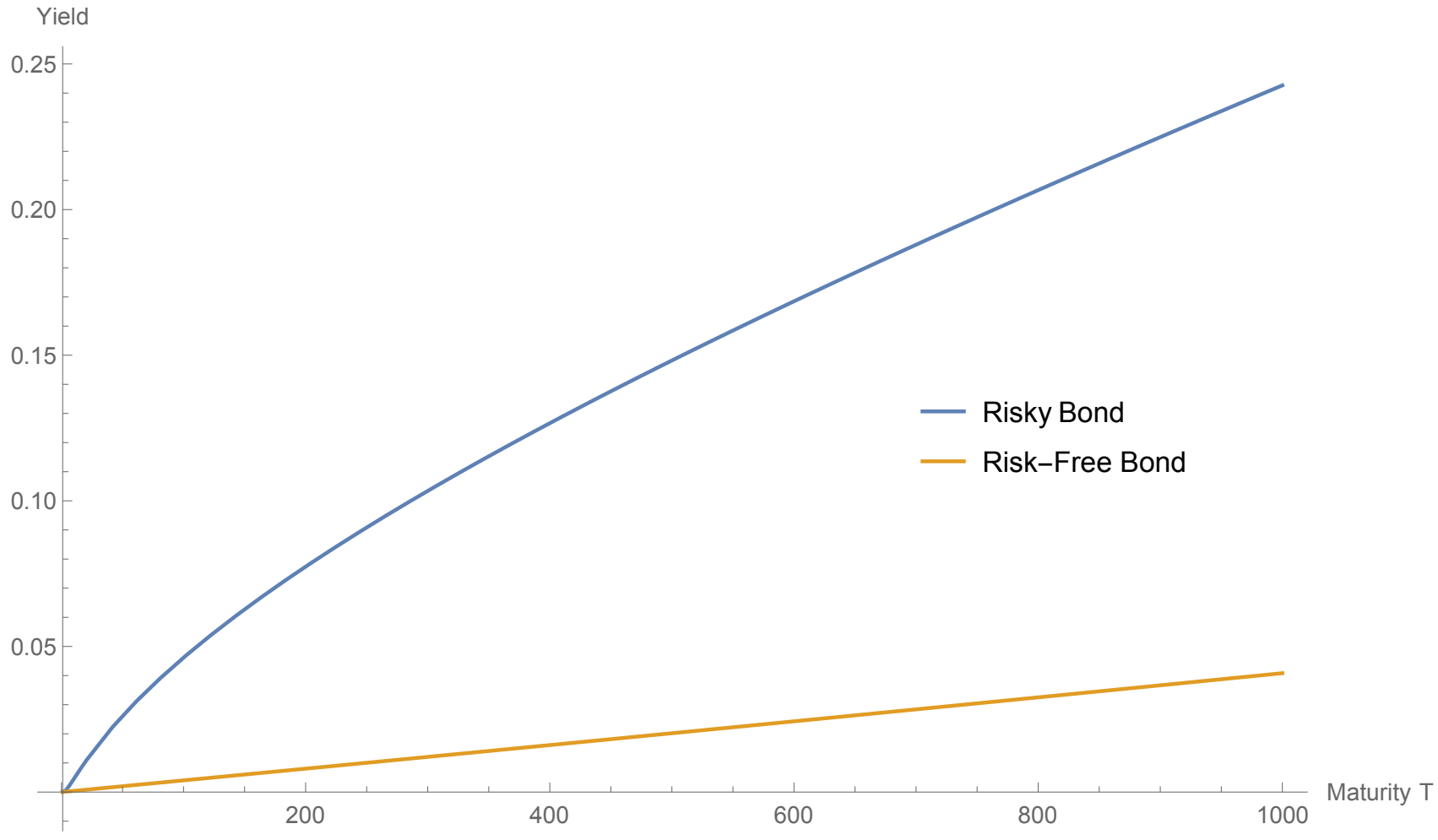
Bond's face value



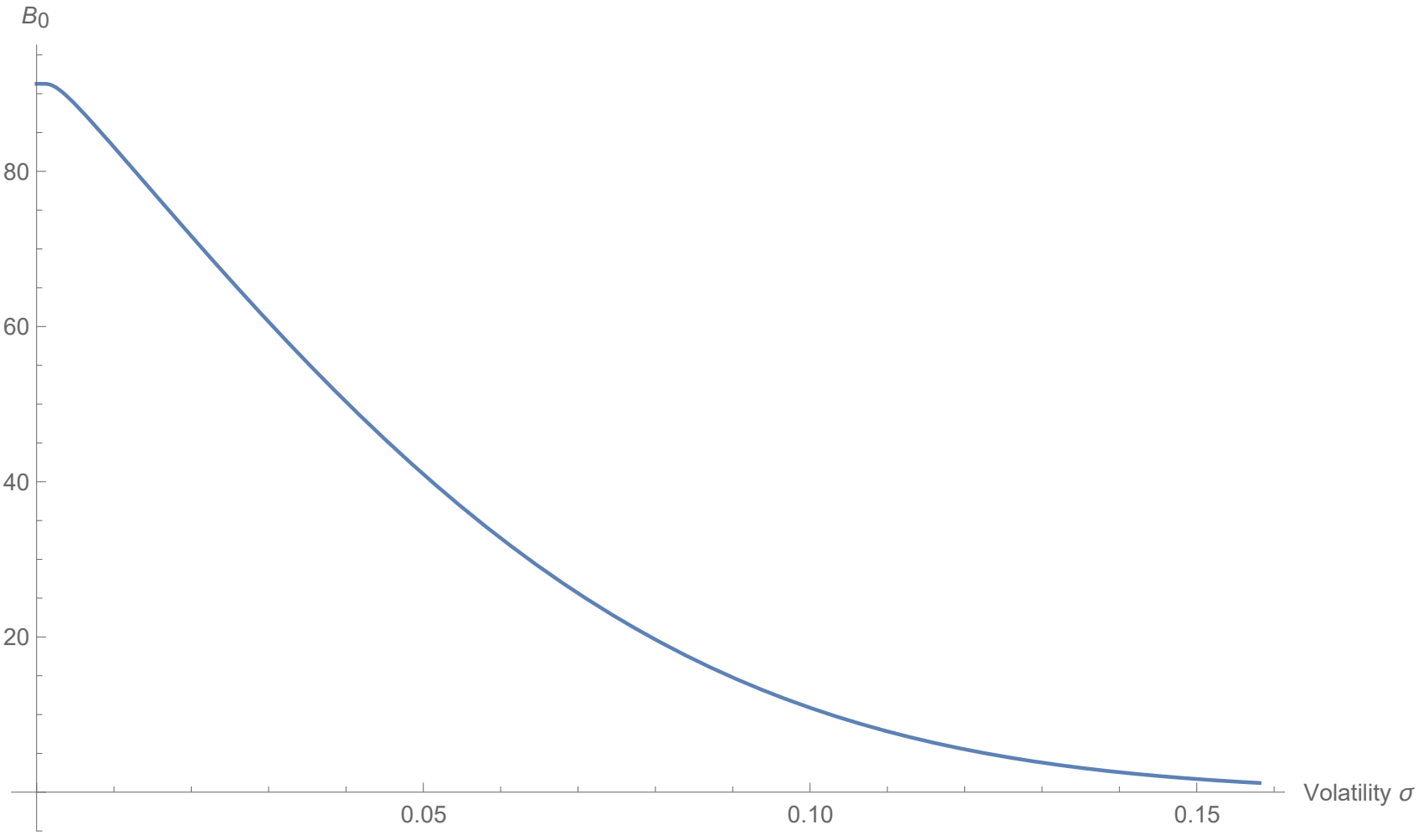
Maturity - Bond price



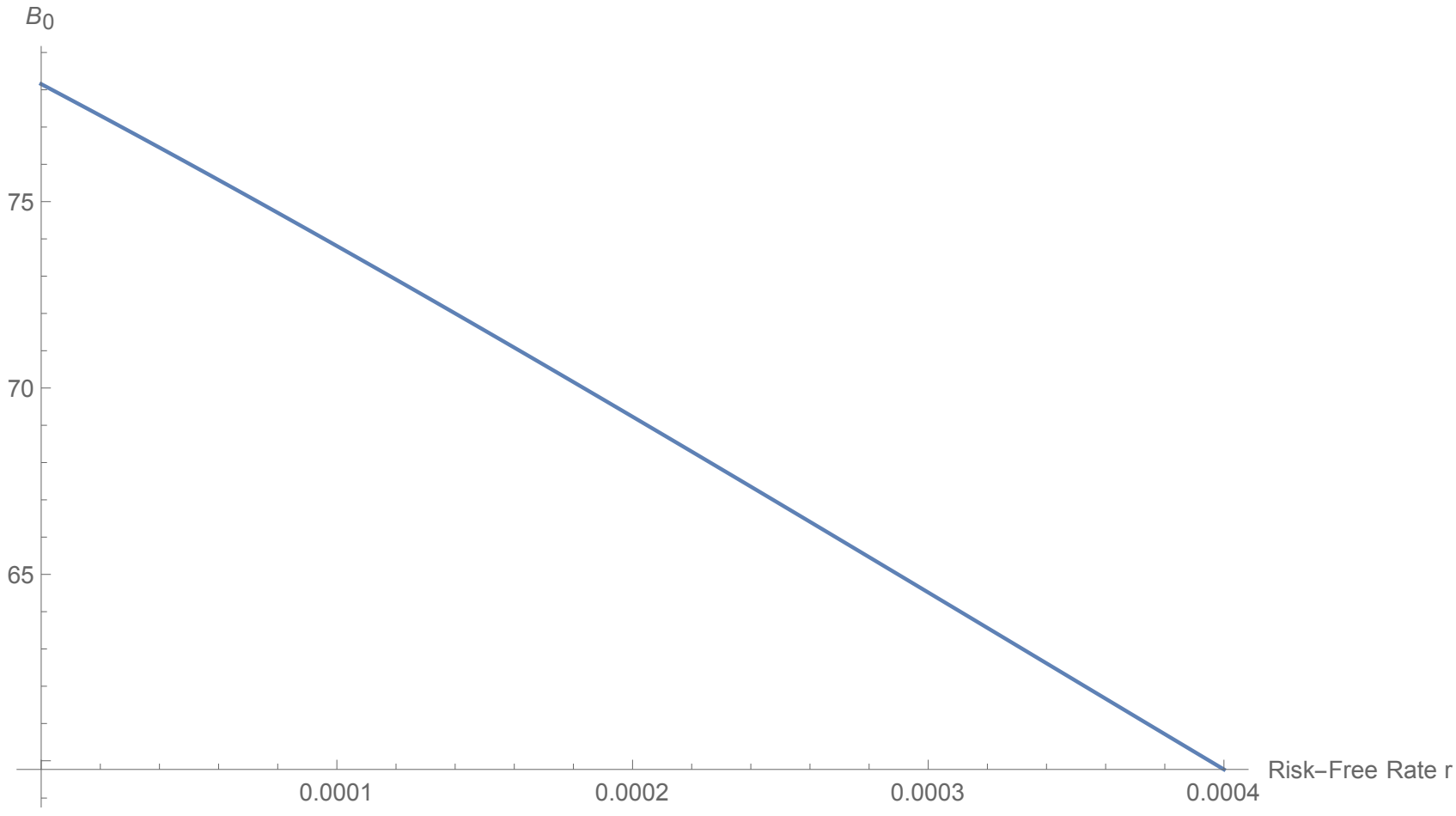
Maturity - Bond yield



Volatility



Risk-free rate



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Structural vs. Reduced-Form

- First structural models were developed when data was scarce.
- "Reduced form" approach: predict default based on variety of signals and datasets, and different methods (e.g., logistic regression).
- Will structural models become outdated as more and more data is available?

Further work after Merton

- Relaxing assumptions requires a lot of additional research work, e.g.:
  - Bankruptcy before T: Black and Cox 1992.
  - Coupon bond: Geske 1992.
  - Optimal capital structure: Leland 1994.
  - Unobservable firm value: Duffie and Lando 2001...
- Models get better over time, but one has to be patient.

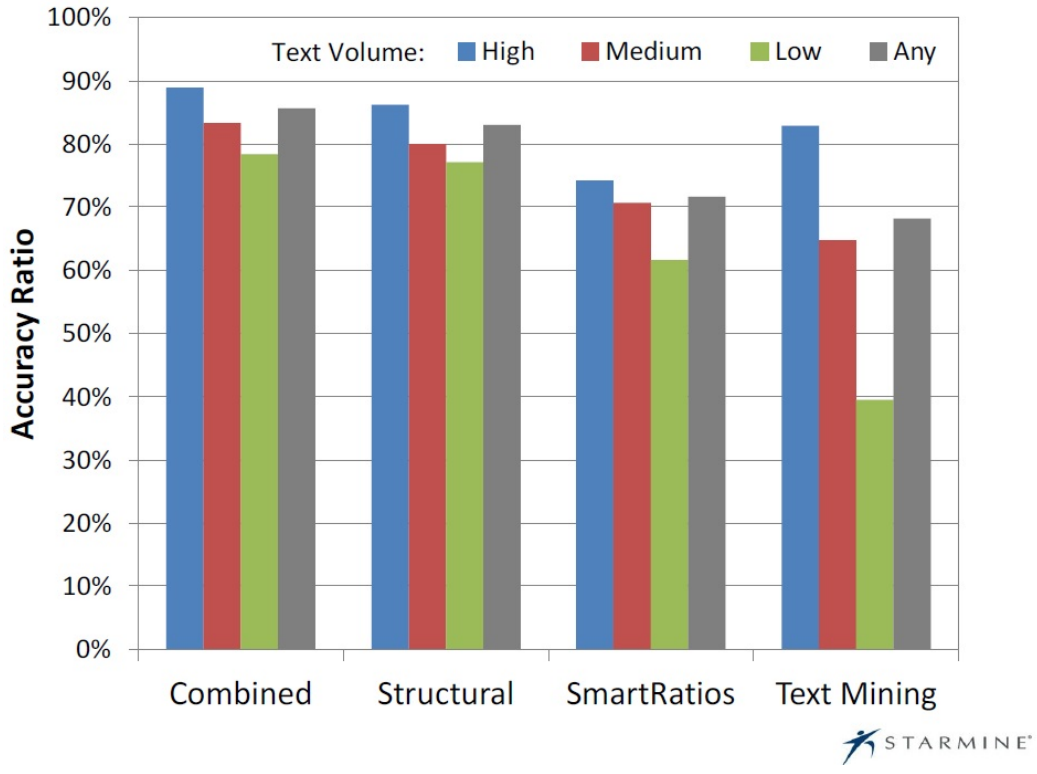
Strengths of structural models

- Explicit assumptions, interpretability.
- Realistic predictions.
- Addresses endogeneity / selection bias issues:
  - Company wants to issue new bond.
  - How costly is it to increase leverage?
  - One can look at the relation between leverage and yield in the data.
  - Relation could be very weak, or even negative. Why?

Weaknesses of structural models

- Throw away a lot of information. In Merton 1974 only the stock price is used.
- Model puts a lot of structure on the data, uses many assumptions.
- Can one do better by adding more data, more flexibly?
- Example: textual analysis on companies' annual filings.

Example



Not easy to beat the structural model... Why?

Market efficiency

- Stock prices already include a lot of data.
- In particular, the info from annual filings is already in the stock price!
- Using signals already incorporated in the stock price not very useful.

When is extra data likely to add value?

- In Merton's model all the info to price the bond is in the stock price.
- In reality, some relevant info for the bond may not be in the stock price, e.g.:
  - Info on LGD (irrelevant for shareholders).
  - Info about "jumps" to the default state.
- Finance theory offers useful guidance to the data scientist here.

Conclusion

- Structural models are an ambitious way to process data.
- Market prices already contain a lot of information.
- Focus should be on alternative data sources unlikely to be incorporated in market prices.
- Key condition here is that the firm has a publicly traded stock.
- Otherwise one needs to aggregate data from many sources without the help of prices $\Rightarrow$ credit scoring, next session.

Thank you!